SDS 320E – Project Proposal

Answer questions in the boxes provided using complete sentences in single-spaced, 12-point font.

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Pilot Dataset

1. (10pt) Enter *n=10* observations in the table below (real observations if you are proposing to use existing data or simulated observations if you are proposing to collect your own).
   - Replace each column name with a short but descriptive variable name.
   - Your "Row_ID" should be a unique value that helps identify each observation in your sample (it can be as simple as "1, 2, 3,…" or names/labels for each sampling unit).

Team	Three-Pointers Attempted Per Game	Three-Point Shooting Efficiency	Win Percentage
2019-20 Bucks	38.082	Low	0.767
2019-20 Raptors	36.986	High	0.736
2019-20 Celtics	34.542	Low	0.667
2019-20 Pacers	28.027	Low	0.616
2019-20 Heat	35.397	High	0.603
2019-20 76ers	31.603	High	0.589
2019-20 Nets	38.139	Low	0.486
2019-20 Magic	32.247	Low	0.452
2019-20 Hornets	34.323	Low	0.354
2019-20 Wizards	32.569	High	0.347

Project Title and Research Questions

2. (1pt) What is the title of your proposed research project? This should be a short sentence or phrase in plain language that describes your project topic and your response variable.

Do teams that shoot more three-pointers win more games in the NBA?

3. (2pt) State the two research questions, one for each explanatory variable's relationship with the response, that you are going to attempt to answer with your analysis. It should be clear which variable is the response and which is the explanatory in each question.

RQ1 – Does the number of three-pointers attempted per game correlate with a team's win percentage?

RQ2 – Does a higher three-point shooting percentage lead to a higher win percentage?

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Motivation for Studying this Topic

4. (5pt) Briefly describe your motivation for choosing this research topic. Cite in APA format at least one source that relates to the relationships you are investigating and briefly describe how your study will confirm, refute, replicate, or expand on it. Your source can be academic or non-academic (articles, news reports, etc. are allowed).

The modern NBA looks radically different from even a decade ago, largely due to the rise of the three-point shot. What was once seen as a situational strategy has evolved into a fundamental pillar of basketball, with teams building entire offensive systems around perimeter shooting. As a player who relies heavily on the three-point shot, I'm curious to examine whether the data supports this dramatic shift in strategy. This research explores the relationship between three-point shooting and team success to evaluate if the league's embrace of the three-pointer is backed by winning basketball[1].

[1] Grimes, P. J. (2024, October 30). The NBA has a 3-pointer problem. USA Today. https://ftw.usatoday.com/2024/10/nba-three-pointers-record-attempts-problem

Sampling and Data Collection Plan

5. (2pts) Who or what will be the individual sampling units for your study? Describe what the observations (not variables) in your dataset will be. **Be specific.**

The individual sampling units for my study will be NBA teams over multiple seasons (2019-2020 season to 2023-2024 season). Each observation represents a team's three-point statistics and win percentage from a single season.

6. (2pt) What is your population of interest (the population will you be making an inference about)? **Be specific**. *Remember: your population must be at the same measurement level as your sampling units!*

The population of interest includes all teams that have competed in the NBA.

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7. (5pt) Describe the process you will use to collect your data. **Be specific.**
   - If you will be collecting your own data, describe exactly where and how you will get your responses. **Include a link to your designed survey if collecting data online.**
   - If you are using existing data, cite the specific source(s) and briefly explain how you plan to extract the relevant variables.

The Basketball Reference website (https://www.basketball-reference.com) and NBA.com provide free access to extensive NBA statistics, including team-level data on three-point attempts, three-point efficiency, and win percentage. These databases contain a variety of statistical measures, such as advanced metrics, shooting splits, player tracking data, and historical trends from past seasons. The statistics are compiled and updated regularly using official NBA game logs, ensuring accuracy and consistency. I will collect data spanning from the 2019-2020 to 2023-2024 seasons to analyze trends in three-point shooting and its impact on team success.

8. (5pt) Will your dataset be considered a random sample from your population of interest? If so, explain why your sample will be representative of your population. If not, describe what limitation this may place on your results.

My sample will not be a random sample from my population of interest because it will only contain NBA teams that participated in the league during the selected seasons. Since the dataset includes all teams from these seasons rather than a randomly chosen subset, external factors such as injuries, coaching strategies, and player trades may impact the results. Also, every team has a new team every single season, so there will never be 2 observations with the same people. Additionally, teams with different playing styles or roster compositions may skew the analysis, as some teams may naturally emphasize three-point shooting more than others. This means my findings will primarily reflect trends within this specific time frame and may not generalize about future NBA seasons or historical eras with different styles of play.

Proposed Measures

9. (3pt) What will be your **response variable**? Include a description of exactly how it will be measured, including units (if applicable), and the specific range of values you expect to get in the your sample.

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The response variable will be the percentage of games a team wins in a season, calculated as (Wins / Total Games) and ranging between 0 and 1. Based on historical data, win percentages typically fall between 0.20 and 0.80, providing a clear measure of team success.

10. (5pt) What will be your **numeric explanatory variable**? Include a description of exactly how it will be measured, including units (if applicable), and the specific range of values you expect to get in the your sample. Then describe in what way you think it is related your response variable. **Be specific** and briefly explain why you think this relationship exists.

The numeric explanatory variable is the average number of three-point shots a team attempts per game, measured in whole numbers. Based on historical NBA data, this value typically ranges from 20 to 45 attempts per game. I expect that teams attempting more three-pointers will generally have a higher win percentage, as modern NBA offenses increasingly prioritize three-point shooting for greater scoring efficiency.

11. (5pt) What will be your **binary categorical explanatory variable?** Include the two possible values it will have and describe in what way you think it is related your response variable. **Be specific** and briefly explain why you think this relationship exists.

The binary categorical explanatory variable is three-point shooting efficiency, categorized into high efficiency and low efficiency. The league median for three-point shooting in the 2023-2024 season was 36.65%, so teams with a three-point shooting percentage of 36.65% or higher will be classified as high efficiency, while those with a percentage below 36.65% will be classified as low efficiency. I expect that high-efficiency teams will generally have a higher win percentage, as making a greater percentage of three-point shots leads to increased scoring efficiency.

Collaborative Institutional Training Initiative (CITI) Report *(if needed)*

If you are collecting your own data from humans, you must complete the CITI training or you will receive a *10 point penalty*. Click here, scroll down to "Human Subjects Research Training" and click "I have never taken a CITI course." Follow the instructions for "Social/Behavioral

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Researchers." If you are collecting data online, complete the "Internet-based Research" for your optional module. If you've completed this training previously, you may upload a completion report dated within the past 3 years.

Paste an image of your completion report below: